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Transcriptome-wide effect of *DE-ETIOLATED1* (*DET1*) suppression in embryogenic callus of *Carica papaya*

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Abstract

Papaya is one of the most nutritional fruits, rich in vitamins, carotenoids, flavonoids and other antioxidants. Previous studies showed phytonutrient improvement without affecting quality in tomato fruit and rapeseed through the suppression of *DE-ETIOLATED-1* (*DET1*), a negative regulator in photomorphogenesis. This study is conducted to study the effects of *DET1* gene suppression in papaya embryogenic callus. Immature zygotic embryos were transformed with constitutive expression of a hairpin *DET1* construct (*hpDET1*). PCR screening of transformed calli and reverse transcription quantitative PCR (RT-qPCR) verified that *DET1* gene downregulation in two of the positive transformants. High-throughput cDNA 3' ends sequencing on *DET1*-suppressed and control calli for transcriptomic analysis of global gene expression identified a total of 452 significant (FDR < 0.05) differentially expressed genes (DEGs) upon *DET1* suppression. The 123 upregulated DEGs were mainly involved in phenylpropanoid biosynthesis and stress responses, compared to 329 downregulated DEGs involved in developmental processes, lipid metabolism, and response to various stimuli. This is the first study to demonstrate transcriptome-wide relationship between light-regulated pathway and secondary metabolite biosynthetic pathways in papaya. This further supports that the manipulation of regulatory gene involved in light-regulated pathway is possible for phytonutrient improvement of tropical fruit crops.

Keywords *DE-ETIOLATED1* · Embryogenic callus · Photomorphogenesis · HpRNAi · RNA-seq

Introduction

Papaya (*Carica papaya* L.), once called the fruit of angels by Christopher Columbus, has a wide range of health benefits (Wall 2006). The fruit is extremely rich in antioxidants that help in promoting better health and preventing cancer (Elgadir et al. 2014). These natural antioxidants include polyphenols, carotenoids, and flavonoids, which are mainly plant secondary metabolites that are also important determinants of food quality, like taste and colour. In developing countries, crop improvement for increasing yield and stress

tolerance is the main priority. However, the improvement of crop nutritional value is also gaining attention. This is because of increasing demand for fruits and vegetables with higher nutrient content for the prevention of non-communicable diseases, such as cancer and cardiovascular disease.

The enhancement of nutritional quality in crop by manipulating specific secondary metabolite pathways has been successfully demonstrated in rice (Ye et al. 2000), tomato (Lewinsohn et al. 2001), and many other plants. On the other hand, the less conventional manipulation of light regulatory pathway also showed surprising effect on simultaneous enhancement of multiple phytonutrients (Davaluri et al. 2005). To date, molecular regulation of light in fruit crop improvement has been demonstrated mainly in tomato, which showed effects on fruit pigmentation due to the enhancement of carotenoid biosynthesis (Enfissi et al. 2010). This is further supported by the increase in flavonoid quercetin and carotenoids (lutein, β -carotene, and zeaxanthin) with reduced phenolic sinapate esters in rapeseed when *DE-ETIOLATED1* (*DET1*) is suppressed (Wei et al. 2009).

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DET1 is a negative regulator or suppressor of photomorphogenesis identified almost 30 years ago (Chory et al. 1989). In the dark, DET1, COP10 and DDB1 combine to form a stable CDD complex, which is involved in the degradation of transcription factors, namely CCA1 and LHY during the suppression of photomorphogenesis and circadian clock (Lau and Deng 2012). DET1 is also present in human to stimulate ubiquitination and degradation of c-Jun transcription factor (Wertz et al. 2004). However, the molecular mechanism on how *DET1* expression alters secondary metabolism is unclear.

Transcriptomics studies of papaya are limited to root (Porter et al. 2008), flower (Urasaki et al. 2012), and fruit (Fabi et al. 2010, 2014). To our knowledge, this is the first transcriptome analysis of transformed papaya embryogenic callus, compared to previously reported wildtype embryogenic callus (Jamaluddin et al. 2017b). In this study, the endogenous photomorphogenesis regulatory gene, *DET1* was suppressed in papaya embryogenic callus to study its effect on genome-wide expression. We are interested in whether *DET1* suppression could alter the expression of genes involved in secondary metabolite biosynthetic pathways similarly to tomato and rapeseed. Transcriptomics analysis was performed using embryogenic calli as a proof-of-concept experiment to inform future crop improvement strategy in papaya. RNA-seq analysis of *DET1*-suppressed and control calli based on 3'mRNA sequencing (Jamaluddin et al. 2017a) was performed to identify differentially expressed genes (DEGs).

DEGs from *DET1* suppression were found to be involved in various biological processes, especially response to oxidative stress, hormone and chemical stimuli, secondary metabolite biosynthesis, and plant defence. This study is the first attempt which demonstrated the feasibility of using RNAi gene silencing approach to alter secondary metabolite pathways in papaya. The results from this study also supports the relationship between photoregulated pathway and secondary metabolite biosynthesis.

Materials and methods

Plant material

Three-month-old green fruits of papaya var. 'sekaki' were harvested from an experimental plot at Universiti Kebangsaan Malaysia. The seeds were surface sterilised for 5 min using 20% sodium hypochlorite containing tween 20 and rinsed three times with sterile distilled water. Zygotic embryos were excised from white immature papaya seeds (Fig. S1) for *Agrobacterium*-mediated transformation.

Establishment of hairpin DET1 (hpDET1) construct

To generate RNAi construct for suppression, 161 bp fragments of conserved *DET1* sequence were generated by RT-PCR using primers, CpDET1_F (5'-CACCGAGGTGCATTAAGAACAAAGC-3') and CpDET1_R (5'-TGCCGGTCC AATGCAGAAAT-3'). The addition of 4 bp nucleotides (CACC) to the forward primer allows the PCR product to be directionally cloned into TOPO vector to generate an entry clone (dTOPO_DET1). The resulting PCR fragments were cloned into Gateway pENTR/dTOPO cloning vector carrying two recombination sites (*attL1* and *attL2*) for LR clonase reaction into destination vector pHellsgate12 containing two *attR1-attR2* cassettes (Helliwell and Waterhouse 2003). Constitutive expression of hairpin DET1 (hpDET1) driven by 35S CaMV promoter suppresses endogenous *DET1* expression. Schematic diagram for the construction of plasmid pHellsgate12_hpDet1 is shown in Fig. S2.

Transformation of *Agrobacterium tumefaciens*

For transformation, 1 µL of pHellsgate12_hpDET1 binary vector was transformed into 50 µL *A. tumefaciens* strain GV3101 via the electro-transformation method. The mix was incubated for 5 min, transferred into a multipipetator (Eppendorf) and subjected to 2.5 kV voltage along the pulse length of 5 ms, before adding 500 µL Luria–Bertani (LB) medium immediately and incubated for 4 h at 28 °C on orbital shaker at the speed of 30 rpm. Colony PCR amplification and antibiotic selection validate the uptake of pHellsgate12_hpDET1 plasmid in transformed agrobacterium.

Agroinfiltration of immature zygotic embryo

The culture of verified transformed agrobacterium was grown overnight in LB broth at 28 °C with swirling (200 rpm) until OD₆₀₀ of 0.4–0.6. Immature zygotic embryos of papaya were infiltrated with agrobacterium culture for 40 min under vacuum and left to be immersed in the culture for another 60 min. Then, agroinfiltrated zygotic embryos were dried on sterile filter paper before co-cultivation on MS medium (Murashige and Skoog 1962) supplemented with 200 µM acetosyringone for 16 h in the dark (Fig. S3). The zygotic embryos were then transferred to MS medium maintained in a controlled growth chamber (CU-22L, Percival Scientific) at 25 ± 1 °C in the dark. A week later, the infected zygotic embryos were cultured in MS medium supplemented with plant growth regulators: 1.5 mg L⁻¹ 2,4-dichlorophenoxy acid (2,4-D), 0.5 mg L⁻¹ α-naphthaleneacetic acid (NAA), and 2.25 mg L⁻¹ benzylaminopurine (BAP) and 1 mg L⁻¹ kinetin, for callus induction (Sun et al. 2011).

with addition of 300 mg L⁻¹ cefotaxime to prevent excessive agrobacterium growth. Regeneration media consist of 0.02 mg L⁻¹ NAA and 0.1 mg L⁻¹ BAP. All the basal media comprise MS medium, 3 g L⁻¹ gelrite and 30 g L⁻¹ sucrose, with pH adjusted to 5.7–5.8. Zygotic embryos were subcultured on cefotaxime medium every 2 weeks until no further agrobacterium growth. The zygotic embryos which turned into calli were then selected on regeneration media containing 50 µg mL⁻¹ kanamycin.

DNA isolation and PCR screening for putative transformed callus

Total genomic DNA of transformed and non-transformed (control) calli were extracted using Wizard Genomic DNA Purification kit (Promega) according to manufacturer's instruction using a minimum weight of 100 mg. PCR screening of the survived calli on selection medium were validated through PCR reactions using 35S promoter forward primer (5'-ATCTTCGTCAACATGGTGGAGCAC-3') and reverse primer, CpDET1_R (5'-TGCCGGTCCAATGCA GAAAT-3').

RNA extraction, library construction and QuantSeq 3' mRNA sequencing

Detailed descriptions of RNA extraction, library concentration and RNA sequencing were provided in a previous data article describing the sequencing statistics of this study (Jamaluddin et al. 2017a). Briefly, 300 mg of 4-week-old transformed calli at proliferation stage from positive PCR screening was aseptically harvested and frozen in liquid nitrogen. RNA was extracted using TRIzol (Invitrogen) according to manufacturer's instruction. RNA quality and quantity were determined by using NanoDrop spectrophotometer and agarose gel electrophoresis prior to cDNA library preparation.

For cDNA library preparation, 500 ng of total RNA samples were cleaned using DNase I kit according to the Rapid out removal DNA kit instruction (Thermo Scientific) and converted into cDNA by using Maxima First Strand cDNA Synthesis Kit (Thermo Scientific). For RNA sequencing, we used QuantSeq 3' mRNA-Seq Reverse (REV) Library Prep Kit (Lexogen) according to manufacturer's instruction to generate compatible library for Illumina sequencing (Moll et al. 2014). Briefly, library generation was initiated by oligodT priming for first strand cDNA which generated one fragment per transcript. The second strand cDNA was subsequently synthesised using random primers. Illumina-specific linker sequences were introduced by the primer. The quality of cDNA libraries was determined using a High Sensitivity DNA Assay 2100 Bioanalyzer (Agilent) for quality control analysis. Sequencing of the callus of papaya cDNA library

with 100 bp single end reads was performed using Illumina HiSeq 2500 system at Australian Genome Research Facility (AGRF) according to standard protocols. Raw reads were deposited to NCBI SRA database (SRP075991).

Transcriptome analysis

Raw sequencing reads were processed to remove low quality sequences (QV < 10 of 4-base sliding window) and unknown sequences with 'N' using Trimmomatic (Bolger et al. 2014) to retain reads with minimum length of at least 20 bases long. To quantify transcript abundance, the processed reads were mapped to a modified papaya genome reference version Cpapaya_113 (<http://www.plantgdb.org/CpGDB/>) to include 500 bp extensions from both ends of the CDS sequences with the 'N' removed. The mapping was performed using bowtie2 (Langmead and Salzberg 2012) with stringent "end-to-end" alignment and all other parameters were set to default values according to recommended data analysis workflow by Lexogen (<http://www.lexogen.com/quantseq-data-analysis>). The reads mapped to the indexed reference were quantified using eXpress (Roberts and Pachter 2013) based on Bowtie2 alignment (Langmead and Salzberg 2012) to estimate gene abundance in count per million (CPM). To identify the differently expressed genes (DEGs) in *DET1*-suppressed and control samples, edgeR analysis was performed based on pairwise comparison between group 1 (TC3 and TC5—*DET1*-suppressed) and group 2 (TC1 and TC4—control). Genes showing false discovery rate (FDR) < 0.05 were defined as statistically significant DEGs.

Functional annotation, gene ontology classification and enrichment analysis

Functional annotation information of the reference papaya gene sequences from PLAZA 3.0 followed that of previously described (Jamaluddin et al. 2017b). Gene ontology (GO) analysis was conducted using WEGO database (<http://wego.genomics.org.cn/cgi-bin/wego/index.pl>). Cytoscape software with Biological Network Gene Ontology (BiNGO) plugin (Maere et al. 2005) were used to determine which GO categories are overrepresented in down-regulated and up-regulated genes group. The first step in BiNGO analysis is to identify homologous gene sequences in *Arabidopsis thaliana* through local blast. GO enrichment analysis was based on hypergeometric test with Benjamini and Hochberg FDR using default parameters for adjusted P-value. To identify significantly enriched pathways, we used a web-based tool KOBAS 2.0 (<http://kobas.cbi.pku.edu.cn/home.do>) which incorporates several pathways databases, including KEGG PATHWAY, BioCyc and Panther (Xie et al. 2011). KEGG pathway annotation was performed using KEGG Automatic Annotation Server (KAAS) Ver. 2.1 (<http://www.genome.jp/>)

tools/kaas/, updated April 3, 2015) (Moriya et al. 2007) based on genes data set with additional plant species (ath, aly, bna, cit, tcc, gmx, fve, csv, pop, vvi, sly, nta, osa, zma), GHOSTX as search program, and SBH (single-directional best hit) as assignment mode.

Reverse transcription quantitative PCR (RT-qPCR) analysis

RT-qPCR analysis was performed for *DET1* and 6 other selected genes (Table 2) to estimate the validity of RNA-seq analysis for expression profile analysis. Gene-specific primers (Table S1) for phytoene synthase (*PSY*), lycopene epsilon cyclase (*LCY*), chalcone synthase (*CHS*), late elongated hypocotyl (*LHY*), phenylalanine ammonia-lyase (*PAL*), and oleosin1 (*OLE*) were designed using online Primer-Blast software (<http://www.ncbi.nlm.nih.gov/tools/primer-blast/>) and synthesised commercially (First BASE Laboratories).

Each reaction was performed using 50 ng of first-strand cDNA and 300 nM of each primer in a 10 µL reaction using Luminaris Color Hi Green qPCR Master Mix (Thermo Scientific) in an iQTM5 real-time PCR detection system (Bio-Rad) with iQTM5 Optical system software (Bio-Rad). The cycling conditions were as follows: 50 °C for 2 min, initial denaturation at 95 °C for 10 min, followed by 40 cycles of 15 s at 95 °C, 30 s at 60 °C, and 30 s at 72 °C. Following the final cycle, melting curve analysis was performed to examine the specificity in each reaction (absence of primer dimers and other nonspecific products). SYBR Green fluorescence of the generated products was continuously monitored throughout the temperature ramp from 65 to 95 °C. The temperature rose in 0.5 °C increments every cycle. A single melt peak for each reaction confirmed the identity of each PCR product. Each assay included a no-template control for every primer pair. Standard curve analysis was performed with five dilution points to ensure similar efficiency for each primer pair. Relative quantification of gene expression was calculated and normalised using housekeeping gene, actin (*ACT1*) based on $2^{-\Delta\Delta C_t}$ method (Livak and Schmittgen 2001).

Results

Agrobacterium-mediated transformation of papaya embryogenic callus

Immature zygotic embryos from young green papaya fruits (Fig. S1) were isolated for agrobacterium transformation with pHellsgate12_hpDET1 binary vector (Fig. S2) for the suppression of endogenous *DET1* expression. After transformation, the explants were cultured on callus induction medium with cefotaxime for 3 weeks before selection

on regeneration medium with kanamycin for transformed embryogenic callus (Fig. S3). A total of seven kanamycin-resistant calli were identified with the presence of hpDET1 construct through PCR screening with an expected amplicon size of 658 bp compared with no amplification in control non-transformants (Fig. S4). There was no apparent phenotypic difference among the transformants. All kanamycin-resistant transformants appeared to be more compact and brownish in comparison to yellowish friable control non-transformants (Fig. S5). Proliferation efficiency and callus growth were observed to be much lower in transformants than the controls perhaps due to the detrimental effects of acetosyringone and antibiotic selection.

Analysis of *DET1* expression by RT-qPCR

To verify that *DET1* was down-regulated by hpDET1 construct, positive transformants were further analysed by RT-qPCR analysis. The results showed that *DET1* expression was successfully down-regulated in two positive transformants (TC3 and TC5) compared to controls (NT6 and NT1), while some transformants (TC1 and TC4) showed similar level of *DET1* expression to wildtype control (Fig. 1). For the comparison on the effect of *DET1* suppression on transcriptome-wide expression, transformants which showed downregulation of *DET1* expression (TC3 and TC5) were chosen for comparison with transformants that were not suppressed (TC1 and TC4) as control.

RNA-seq analysis

High throughput cDNA 3' ends mRNA sequencing (RNA-seq) was performed on Illumina HiSeq 2500 platform for two biological replicates of *DET1*-suppressed and control

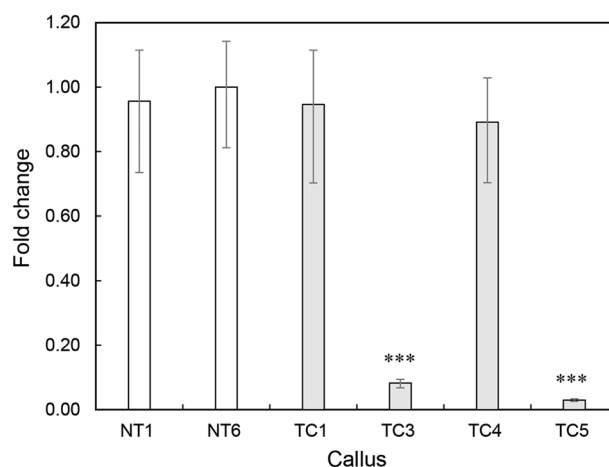


Fig. 1 RT-qPCR analysis of *DET1* expression in non-transformant (NT) and transformed calli (TC). Error bars represent standard errors from average of three technical replicates; T-test *** $P < 0.001$

transformants. Read processing and mapping statistics were reported previously (Jamaluddin et al. 2017a). The number of raw reads from the four samples range from 4.4 to 7.3 mil reads, and the number reduced to 3.2–5.5 mil reads after processing. This is well above the minimum number (0.625 mil) required for confident estimation of gene expression (Moll et al. 2014) even with mapping percentage of 43–63%. This is due to the advantage of 3' mRNA sequencing in generating only one fragment from each transcript, thus circumventing the phenomenon in which conventional mRNA sequencing reads of long abundant transcripts crowding out short and low abundant transcripts.

Based on read mapping results, transcript abundance was estimated and normalised as count per million (CPM) for 28,072 reference papaya genes (Supplementary Data). The two biological replicates of control (TC1 and TC4) and *DET1*-suppressed (TC3 and TC5) samples showed very high correlation, with Pearson's correlation coefficient (r) of 0.999 and 0.998, respectively. Therefore, mean CPM values from each set of samples were used for further analysis. Distribution of gene expression was tabulated according to different levels of transcript abundance with mean CPM values of zero represent non-expressed genes (Table 1). More genes were found to be absent or expressed in low abundance ($\text{CPM} \leq 5$) in *DET1*-suppressed samples (17,594; 62.7%) compared to control (16,603; 59.1%). Furthermore, at the expense of fewer genes with mid abundance ($5 < \text{CPM} \leq 100$), *DET1*-suppressed samples also showed greater number of genes (1,263; 4.5%) with high abundance ($\text{CPM} > 100$), compared to control (1,175; 4.2%).

Functional analysis of differentially expressed genes (DEGs)

Pairwise comparison between group of *DET1*-suppressed and control calli were performed and a total of 452 genes were identified as DEGs ($\text{FDR} < 0.05$) upon *DET1* suppression and were used for subsequent downstream analysis. Among them, 123 genes were found to be significantly upregulated with

$\text{Log}_2\text{FC} > 1.95$, while 329 were significantly downregulated with $\text{Log}_2\text{FC} < -1.99$ (Supplementary Data).

To understand the relevance of DEGs to different physiological functions based on gene ontology (GO) classification, WEGO functional gene annotation tool was used to classify DEGs according to molecular function, biological process, and cellular component (Fig. S6). Within the biological process category, metabolic process, cellular process, and response to stimulus were the top most represented GO terms for all DEGs. Furthermore, the majority of DEGs have binding and catalytic activities.

Gene ontology enrichment analysis was performed separately for upregulated and downregulated DEGs using BiNGO based on Arabidopsis model. Upregulated DEGs showed significant enrichment in response to oxidative stress, plant cell wall modification, phenylpropanoid metabolic and biosynthetic processes, aromatic compound biosynthetic process, and cellular amino acid derivative biosynthesis process (Fig. 2). Conversely, downregulated DEGs were enriched in response to hormone (abscisic acid and gibberellin) stimulus, response to chemical (organic substance) stimulus, oxylipin, glycosides and sucrose metabolic processes, as well as various developmental processes (Fig. 3).

To further understand the functions of DEGs in biosynthetic pathways, we performed pathway enrichment analysis (Table 3). Upregulated DEGs were enriched significantly in phenylpropanoid biosynthesis, pentose and glucuronate interconversions, chitin degradation II, and homogalacturonan degradation. Conversely, downregulated DEGs were enriched in ethanol degradation I, acetaldehyde biosynthesis I, pyruvate fermentation to ethanol II, fatty acid degradation, and ethanol degradation II. This is depicted by KEGG pathway mapping (Fig. 4), which showed greater number of upregulated DEGs mapped to phenylpropanoid biosynthesis, phenylalanine metabolism, and fructose and mannose metabolism. In comparison, downregulated DEGs were mainly mapped to cutin, suberin and wax biosynthesis, as well as starch and sucrose metabolism. Overall, this indicates the repression of lipid metabolism and the induction of phenylpropanoid metabolism in *DET1*-suppressed calli.

On the other hand, a total of seven upregulated and 42 downregulated transcription factors were found (Fig. S7). Based on the proportion to the number of expressed transcription factors, zinc-finger homeodomain protein (ZF-HD) and growth-regulating factor (GRF) were the two most downregulated transcription factor families. Selected DEGs and non-DEGs are listed in Table 2 according to functional relevance.

Effect of *DET1* suppression on light-regulated pathway genes

DET1 is an endogenous photomorphogenesis regulatory gene and interacts physically with two transcription

Table 1 Distribution of average transcript abundance in control and *DET1*-suppressed calli

Normalised transcript count (CPM)	Transcript abundance	Number of genes	
		Control	<i>DET1</i> -suppressed
0		6,439	7,051
> 0–5	Low	10,164	10,543
> 5–100	Mid	10,294	9,215
> 100–1000	High	1,123	1,189
> 1000	Very high	52	74
	Total	28,072	28,072

biological_process

growth

developmental growth involved in morphogenesis

developmental growth

unidimensional cell growth

cell growth

cellular process

cellular metabolic process

metabolic process

secondary metabolic process

cellular amino acid and derivative metabolic process

cellular amino acid derivative biosynthetic process

cellular biosynthetic process

aromatic compound biosynthetic process

phenylpropanoid biosynthetic process

cellular amino acid derivative metabolic process

phenylpropanoid metabolic process

response to stimulus

response to stress

response to chemical stimulus

cell wall organization or biogenesis

cell wall modification

plant-type cell wall modification

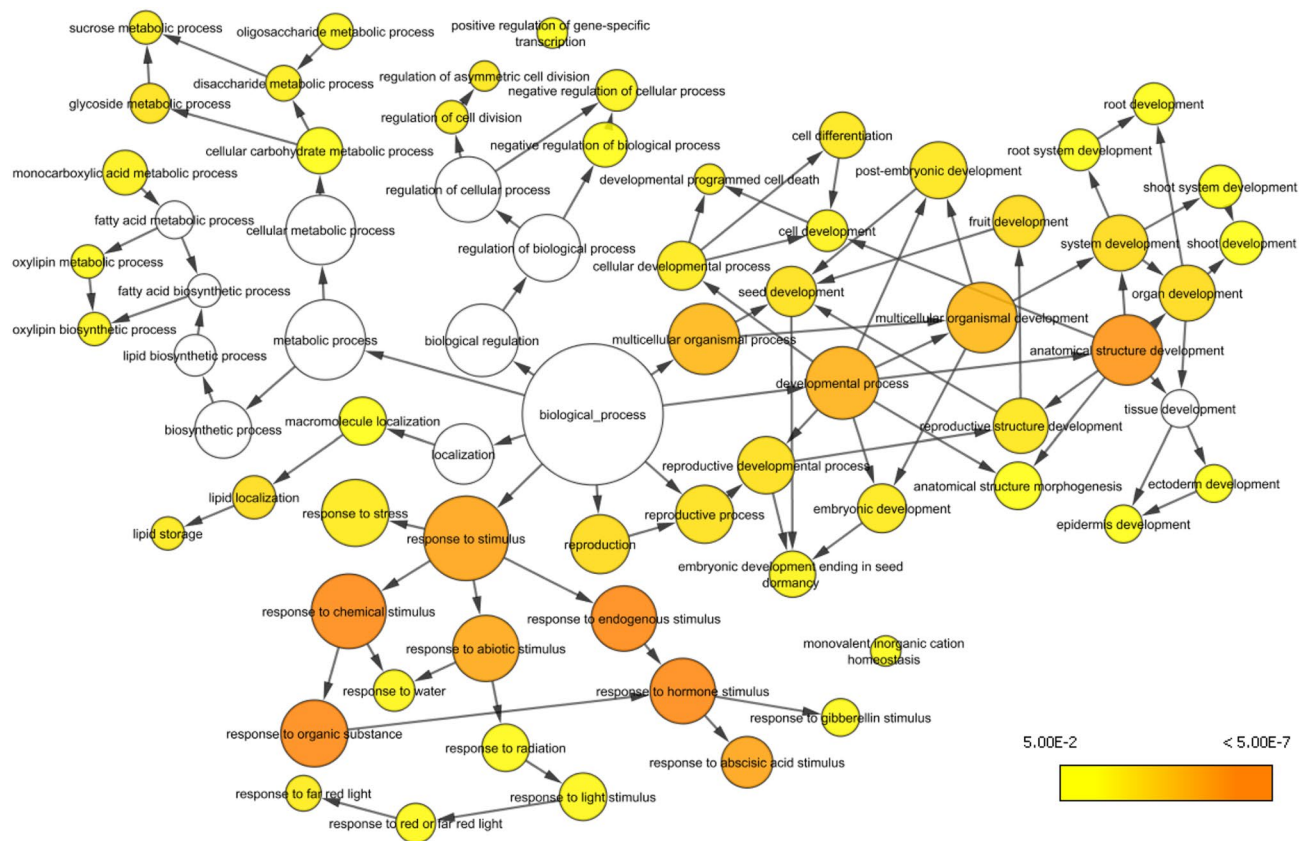
cell wall organization

plant-type cell wall organization

plant-type cell wall loosening

5.00E-2

< 5.00E-7



indicates more significantly overrepresented. White nodes are not significantly overrepresented. The size of a node is proportional to the number of genes in the test set annotated to the corresponding GO category

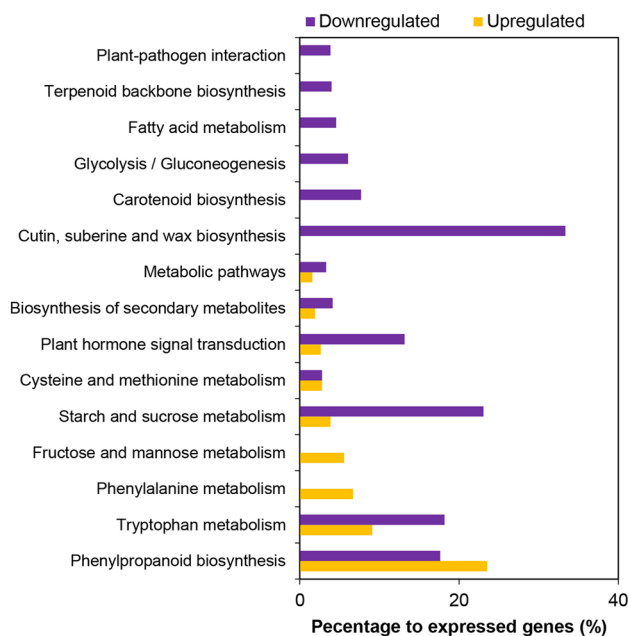


Fig. 4 KEGG pathway mapping of DEGs. Relative percentage of the number of mapped DEGs relative to the number of mapped expressed genes with CPM > 1

factors, CCA1 and LHY to suppress their transcription in the absence of light. COP/DET/FUS proteins are negative regulators of light signalling, which target photomorphogenesis-promoting transcription factors, such as ELONGATED HYPOCOTYL 5 (HY5), for proteasomal degradation. DET1 also interacts physically with DDB1 and COP1 to form CDD complex, and together with CUL4, regulates photomorphogenesis. In this study, the expression level of *FUS9*, *DDB1A/B*, *COP1*, and *CUL4* showed lower expression compared to higher expression of transcription factors *LHY*, *HY5*, and *GI* (Table 2). This is consistent with the role of DET1 as a negative regulator. However, these differences in expression levels were not statistically significant in RNA-seq analysis despite largely showing consistent lower or higher expression between biological replicates of control and *DET1*-suppressed samples (Supplementary Data).

RT-qPCR validation

Based on CPM values, average *DET1* expression was lower in *DET1*-suppressed samples as compared to control despite not being statistically significant in RNA-seq analysis (Table 2). However, qPCR results showed significant down-regulation of *DET1* expression at 0.58-fold (Fig. 5a). To validate RNA-seq analysis, we selected a few candidate genes related to secondary metabolites pathways (flavonoid and carotenoid), such as phenylalanine ammonia-lyase (*PAL*), chalcone synthase (*CHS*), phytoene synthase (*PSY*), and lycopene epsilon cyclase (*LCY*); a gene involved in oilseed

body, oleosin (*OLE*), and a transcription factor related to photoregulated pathway, late elongated hypocotyl (*LHY*). The expression of almost all genes obtained by RT-qPCR were consistent with RNA-seq result with $R^2 > 0.92$, such that *PSY*, *PAL*, *CHS* and *LHY* were upregulated ($FC > 1$), compared to *OLE* and *DET1* which were downregulated ($FC < 1$). *LCY* was found to be upregulated in RT-qPCR compared to RNA-seq analysis which indicated lower expression despite being not significant (Table 2).

Discussion

Immature zygotic embryos (Fig. S1) were used as explants for *Agrobacterium*-mediated transformation because of higher viability and callusing rate. We observed callus induction after only 3 weeks of culture as reported by Fitch and Manshardt (1990), compared to much longer time when using different explants, such as root (Chen et al. 1987) and ovule (Litz and Conover 1981). The first transformation of papaya using *Agrobacterium* was reported by Fitch et al. (1993). In this study, zygotic embryos were infected with *Agrobacterium* strain GV3101 containing pHells-gate12_hpDET1 plasmid (Fig. S2) through bacterial culture immersion with vacuum infiltration (Fig. S3). During co-cultivation, the addition of acetosyringone, a natural wound-induced phenolic compound not toxic to *Agrobacterium*, helped to enhance the virulence of *Agrobacterium* infection, which has been shown to increase the transformation efficiency in papaya (Zhu et al. 2007).

However, there are limitations in *Agrobacterium*-mediated transformation, including low efficiency of transgene transfers into explants, and difficulty in controlling *Agrobacterium* overgrowth and plant regeneration. Cefotaxime inhibits *Agrobacterium* growth by blocking the biosynthesis of cell wall mucopeptide (Mathias and Mukasa 1987). Increasing the concentrations of cefotaxime and antibiotics can potentially inhibit the development of embryogenic callus. The overgrowth of *Agrobacterium* can possibly be remediated by lower culture density and/or reduced duration of co-cultivation from 72 to 24 h, as demonstrated in suspension culture of papaya cultivar ‘Kopoho’ (Carlos-Hilario and Christopher 2015). Nonetheless, despite only 16 h of co-cultivation, we still lost many of the explants to bacterial overgrowth. Only seven positive transformants (Fig. S4) were successfully obtained from over 300 explants in at least three attempts, in which only two embryogenic calli showed *DET1* suppression.

Due to this limitation, we can only perform RNA-seq analysis based on two biological replicates (TC1 and TC4) comparing between *DET1*-suppressed transformants with transformants showing normal *DET1* expression level (TC3 and TC5) as validated by RT-qPCR (Fig. 1). It is common

Table 2 Transcript abundance of selected DEG and non-DEG in *DETI*-suppressed papaya embryogenic callus

Gene ID	CPM value		Log ₂ FC	FDR	Description
	Control	Suppressed			
Cell signalling and transcription regulation					
CP00042G00500	13.8	154.1	3.5	0.000	BR11-associated receptor kinase 1
CP00052G00870	6.8	106.6	4.0	0.000	Auxin-induced protein AUX28
CP00006G03210	9.1	42.7	2.3	0.013	Squamosa promoter-binding-like protein (SBP)
CP00079G00540	8.8	34.4	2.1	0.036	WRKY transcription factor 41
CP00195G00400	4.5	29.5	2.7	0.004	BEL1-like homeodomain protein (E2F/DP)
CP00152G00570	386.7	36.0	−3.4	0.013	Aminocyclopropane-1-carboxylate oxidase (ACO)
CP00183G00300	189.5	33.9	−2.3	0.032	Homeobox-leucine zipper protein ATHB-13 (HD-ZIP)
CP00292G00070	239.9	11.1	−4.4	0.000	Gibberellin-regulated protein
CP00233G00090	101.9	5.4	−4.2	0.000	Auxin-responsive protein IAA33
CP02304G00010	56.4	4.2	−3.7	0.000	Ethylene-responsive transcription factor RAP2-3
CP00055G00520	30.4	4.0	−2.8	0.011	ZF-HD homeobox protein
CP01729G00010	43.8	1.8	−4.5	0.000	Growth-regulating factor
CP00005G00280	77.3	1.1	−5.8	0.000	Protein ABSCISIC ACID-INSENSITIVE 5
CP00008G01780	171.1	0.5	−8.5	0.000	B3 domain-containing transcription factor ABI3
CP01076G00010	12.6	0.3	−4.8	0.002	ZF-HD homeobox protein At4g24660

Table 2 (continued)

Gene ID	CPM value		Log ₂ FC	FDR	Description
	Control	Suppressed			
Cellular homeostasis					
CP00170G00370	434.6	2927.7	2.8	0.001	Non-specific lipid-transfer protein 2
CP00213G00130	121.1	517.0	2.2	0.046	Annexin-like protein RJ4
CP00010G02210	13605.8	9.5	−10.5	0.000	2S seed storage protein 5
CP00010G02170	280.2	0.3	−10.0	0.000	2S seed storage protein 5
Cell wall modification					
CP27379G00020	77.7	688.1	3.3	0.001	Endoglucanase 13
CP00518G00020	35.6	526.7	4.0	0.001	WALLS ARE THIN 1
CP00862G00020	19.4	429.3	4.6	0.000	Putative pectate lyase 11
CP00004G00380	21.9	361.5	4.2	0.000	Putative pectate lyase 14
CP00081G01390	45.2	257.9	2.7	0.017	Expansin-A4
CP00025G01950	32.6	156.7	2.4	0.020	Pectinesterase/pectinesterase inhibitor
CP00047G00480	23.1	155.9	2.9	0.001	Glucomannan 4-beta-mannosyltransferase 9
CP00034G01390	3.7	44.8	3.8	0.000	Expansin-B16
Somatic embryogenesis					
CP00005G01060	1777.9	4.0	−8.7	0.000	Embryonic protein DC-8
CP00058G00890	168.1	2.0	−6.2	0.000	Late embryogenesis abundant protein D-34
CP00106G00030	306.1	0.4	−9.4	0.000	Embryogenic cell protein 40
Stress response					
CP00012G02110	74.4	1137.5	4.1	0.000	Osmotin-like protein
CP00006G03310	216.2	867.2	2.1	0.024	Catalase isozyme
CP00055G00990	396.4	77.9	−2.2	0.028	Germination-specific cysteine protease 1
CP00452G00010	103.4	21.9	−2.1	0.024	Protein DnaJ

Table 2 (continued)

Gene ID	CPM value		Log ₂ FC	FDR	Description
	Control	Suppressed			
CP00021G02220	107.3	16.6	−2.6	0.013	HSP41
CP00029G00510	61.2	8.7	−2.7	0.002	HSP90
CP00085G00040	102.9	6.5	−3.9	0.000	Universal stress protein A-like
Metabolism					
CP00157G00450	4624.4	18953.0	2.1	0.041	Adenosylhomocysteinase
CP02587G00010	129.6	991.3	3.0	0.001	Dirigent protein 20
Phenylpropanoid biosynthesis					
CP00092G01180 ^a	1476.9	6855.9	2.3	0.029	Phenylalanine ammonia-lyase
CP00055G00010	328.0	1269.7	2.1	0.049	Cinnamyl alcohol dehydrogenase
CP00006G02040	158.1	992.3	2.7	0.001	Cytochrome P450 84A1
CP00042G00840	217.7	976.5	2.3	0.020	Lignin-forming anionic peroxidase
CP00006G02030	17.6	74.0	2.2	0.014	Cytochrome P450 84A1
CP00002G03070	8.6	70.0	3.1	0.003	Lignin-forming anionic peroxidase
CP00146G00460	62.5	1.1	−5.5	0.000	1-Cys peroxiredoxin
Flavonoid biosynthesis					
CP00002G02720	628.6	1180.5	1.0	0.741	<i>Trans</i> -cinnamate 4-monoxygenase
CP00200G00270	351.6	805.5	1.2	0.529	Caffeoyl-CoA <i>O</i> -methyltransferase
CP00107G00140	76.7	668.2	3.2	0.000	Amorpha-4,11-diene 12-monoxygenase
CP00146G00360	37.6	138.5	2.0	0.049	Flavonol synthase/flavanone 3-hydroxylase
CP00246G00140	1.2	3.9	1.7	0.798	Dihydroflavonol 4-reductase (DFR)
CP00059G01060 ^a	2.0	2.7	0.3	1.000	Chalcone synthase (CHS)
Carotenoid biosynthesis					
CP00107G01070	71.3	106.1	0.6	1.000	Carotene 3-hydroxylase
CP00119G00720 ^a	31.1	50.3	0.7	0.925	Phytoene synthase 2

Table 2 (continued)

Gene ID	CPM value		Log ₂ FC	FDR	Description
	Control	Suppressed			
CP00077G01030	9.0	24.4	1.6	0.580	9- <i>cis</i> -Epoxycarotenoid dioxygenase NCED5
CP30463G00020	1.9	8.4	2.2	0.157	Anthocyanin 3'- <i>O</i> -beta-glucosyltransferase
CP00117G00690	4.5	6.2	0.5	1.000	Zeta-carotene desaturase
CP00028G01150 ^a	4.9	2.3	-1.2	0.965	Lycopene epsilon cyclase (LCY)
CP00103G00870	11.1	0.5	-4.1	0.001	Xanthine dehydrogenase/oxidase
Oleosin					
CP00013G02360 ^a	1178.9	20.2	-5.7	0.000	Oleosin 14.9 kDa
CP01077G00030	21.0	2.8	-2.8	0.077	Oleosin 5
CP00110G00140	139.5	1.9	-6.0	0.000	Oleosin 18 kDa
CP00108G00590	499.9	0.8	-9.5	0.000	Oleosin 18.2 kDa
CP01868G00020	356.7	0.4	-9.2	0.000	Oleosin 16 kDa
CP00013G00530	12.5	0.4	-4.7	0.000	Oleosin 16 kDa
Light-regulated pathway					
CP00057G00750 ^a	43.7	57.2	0.5	1.000	LHY
CP00017G00600	70.6	56.4	-0.2	1.000	DNA damage-binding protein 1 (DDB1A)
CP00166G00010	54.2	27.5	-0.9	0.824	Cullin-4B (CUL4)
CP00115G00100	6.5	8.0	0.4	1.000	Transcription factor HY5
CP00021G02180 ^a	9.2	6.6	-0.4	1.000	DET1
CP00139G00590	10.0	3.7	-1.2	0.729	DNA damage-binding protein 1 (DDB1B)
CP00132G00330	5.8	3.2	-0.7	1.000	Protein ligase COP1
CP00026G00830	2.0	2.5	0.5	1.000	GIGANTEA (GI)
CP00119G00620	3.8	1.9	-0.9	1.000	COP10/FUS9
CP32775G00010	0.8	0.4	-0.7	1.000	DNA damage-binding protein 1 (DDB1)

^aGenes chosen for RT-qPCR validation

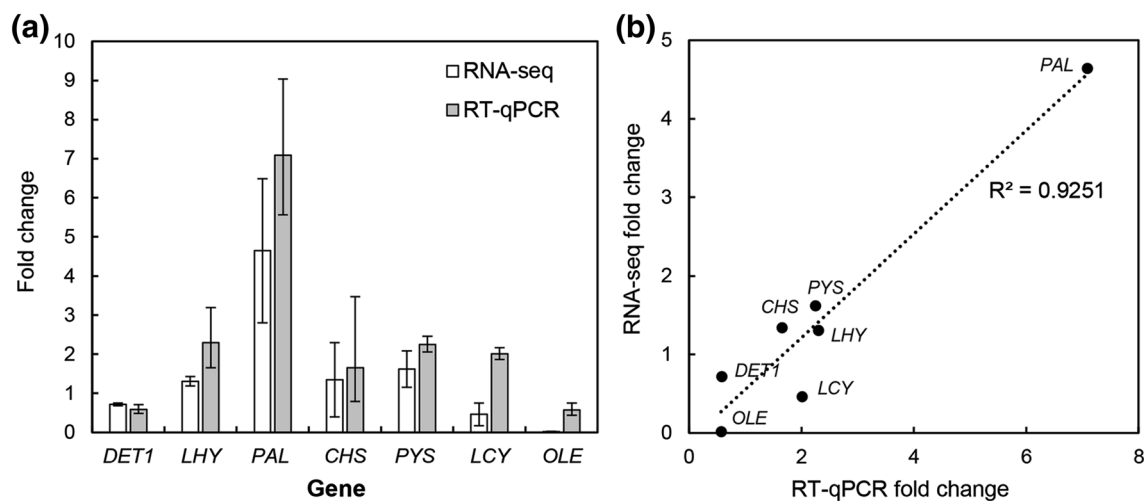


Fig. 5 RT-qPCR validation of RNA-seq gene expression analysis on selected genes. **a** Fold change values from RNA-seq and RT-qPCR of *DET1*-suppressed transformants relative to control. Error bars represent standard errors from average of two biological replicates. **b** Correlation analysis between results from RNA-seq and RT-qPCR

that not all positive transformants with the transgene are *DET1*-suppressed through similar RNAi approach with varied levels of *DET1* expression as seen previously in different transgenic lines of *B. napus* (Wei et al. 2009) and in different stages of transgenic tomato fruit ripening (Enfissi et al. 2010). Comparison between transformants was made to account for experimental conditions of the transgenic calli, in that transformants showing normal *DET1* expression should be a better control than wildtype embryogenic calli to ascertain the genome-wide effect of *DET1* suppression without the confounding factors of transformation and selection.

From RNA-seq analysis, 452 DEGs (FDR < 0.05) were identified with majority of downregulated genes. Gene set enrichment analysis showed that most of the downregulated genes were involved in response to organic substances, hormone, chemical, organic, and endogenous stimuli. This might reflect differential response of *DET1*-suppressed transformants compared to control towards in vitro condition during callus induction and transformant selection with plant growth regulators, kanamycin, and cefotaxime.

Osmotin-like protein and catalase isozyme are among the two highly upregulated genes involved in stress response (Table 2). There are also stress-related proteins which were downregulated, especially those involved in protein folding. Interestingly, most of the DEGs involved in cell wall modification were upregulated in contrast to highly downregulated genes in somatic embryogenesis. Furthermore, genes related to ethylene, gibberellin, and abscisic acid (ABA) were downregulated compared to the upregulated genes related to brassinosteroid and auxin (Table 2). *DET1* suppression has also resulted in the downregulation of four growth-regulating factors (GRFs) important for cell proliferation (Fig.

S7). This might explain why we failed to regenerate plantlets from viable transformants with slowly proliferating calli.

Almost all six oleosin (*OLE*) genes in *DET1*-suppressed calli showed decrease in expression level (Table 2), consistent with the repression of lipid metabolism (Table 3; Fig. 3). Previous study in *Brasica napus* and *Arabidopsis* reported that *OLE* is highly expressed in seed for the biosynthesis of oil seed body to maintain morphology and stabilise the fatty acids storage (Katavic et al. 2006; Siloto et al. 2006). *OLE* suppression resulted in the delay of *Arabidopsis* seed germination process due to changes in the size of the oil seed body with disrupted structure of organelles (Siloto et al. 2006). It is unclear on the role played by oleosin in embryogenic callus and why *DET1* suppression resulted in the downregulation of all oleosin genes in papaya. Oleosin mRNA was shown to be ABA-inducible (Keddie et al. 1994) and hence their reduced expression might be related to the downregulation of ABA signalling pathway (Fig. 3). Furthermore, seed storage proteins were also found to be downregulated in *DET1*-suppressed calli (Table 2).

DET1 can function as a transcriptional co-repressor which suppressed de-etiolated process during seedlings development. This suppression is mainly due to the ubiquitination and degradation of photomorphogenic transcription factors, such as HY5, LHY and CCA1 (Lau and Deng 2012). Previously, protein level of HY5 in *det1* mutant was shown to increase although *DET1* does not interact directly with HY5 (Osterlund et al. 2000). *DET1* directly interacts with SINAT5 E3 ligase for the degradation of transcription factors, CCA1 and LHY, which are the component for circadian clock in *Arabidopsis*. However, no CCA1 ortholog is found in papaya. Through interaction with CCA1 and LHY, *DET1* plays the role of a suppressive pacemaker for

Table 3 Pathway enrichment analysis for upregulated and downregulated DEGs

Pathways	Database	Input no.	Background no.	P value	Corrected P value
Upregulated					
Phenylpropanoid biosynthesis	KEGG	10	157	2.19E–08	1.04E–05
Pentose and glucuronate interconversions	KEGG	6	81	7.87E–06	4.69E–04
Chitin degradation II	BioCyc	2	13	0.0011	0.0149
Homogalacturonan degradation	BioCyc	3	62	0.0011	0.0157
Downregulated					
Ethanol degradation I	BioCyc	3	9	0.0003	0.0127
Acetaldehyde biosynthesis I	BioCyc	3	13	0.0006	0.0127
Pyruvate fermentation to ethanol II	BioCyc	3	13	0.0006	0.0127
Fatty acid degradation	KEGG	4	41	0.0007	0.0127
Ethanol degradation II	BioCyc	3	19	0.0016	0.0227

the Arabidopsis flowering process. Recently, *det1* mutants were shown to exhibit an increase in the activity of a flowering transcription factor Gigantea (GI) (Kang et al. 2015). This is consistent with DET1 as a negative regulator with the upregulation of *LHY*, *HY5*, and *GI* upon *DET1* suppression, in which *LHY* is shown to be upregulated through RT-qPCR (Fig. 5). Interestingly, other components of CDD complex (CUL4 and DDB1) also showed lower expression with the suppression of *DET1* despite being not statistically significant.

On the other hand, genes encoding cell wall modifying enzymes, such as endoglucanase, pectate lyase, pectinesterase, and expansin (Table 2), were significantly upregulated and supported by BiNGO analysis (Fig. 2). This is akin to the effect of cell elongation during etiolation, which further supports that DET1 is a negative regulator of light regulatory pathway in papaya. Changes in the expression of cell wall modifying enzymes are not mentioned in fruit-specific *DET1* downregulation which focused more on the consequence in metabolite composition (Enfissi et al. 2010). It is unclear whether *DET1*-suppressed tomato fruit ripening was affected.

The increase of flavonoids and carotenoids was reported in *DET1*-suppressed tomato fruit which is believed to be attributed by altered plastid biogenesis (Enfissi et al. 2010). However, plastid development is premature in papaya embryogenic callus and no DEGs related to plastid development was detected in this study. The effect of *DET1* suppression is therefore likely to be an indirect consequence of altered expression in some of the transcription factors. Of the 67 MYB and 79 bHLH expressed (CPM > 1) transcription factors (Supplementary Data), which are implied in the regulation of flavonoid pathway (Allan et al. 2008), only two bHLH transcription factors (SPATULA, CP00021G01730; bHLH67, CP00057G00820) with no known relevance to flavonoid biosynthesis were downregulated. One possibly ABA-responsive MYB-related transcription factor

(MYB1R1, CP00013G02350) was downregulated out of 39 other expressed members. Detailed roles of the 49 dysregulated transcription factors (Fig. S7) due to the suppression of *DET1* are yet to be elucidated with further functional studies.

Despite no clear dysregulation of transcription factor involved in flavonoid pathway, phenylalanine ammonia-lyase (PAL), which catalyses the first step of the phenylpropanoid pathway, was found to be highly upregulated upon *DET1* suppression (Table 2; Fig. 5). PAL is the main enzyme that produces the precursor for various phenolic compounds in the biosynthesis of most plant secondary metabolites, especially flavonoids and lignin. Various other enzymes involved in phenylpropanoid and flavonoid biosynthesis were also found to be expressed at higher transcript abundance albeit some are not statistically significant (Table 2). Based on DEGs, there were more downregulated than upregulated genes in metabolic pathways (Fig. 4). However, in general there were more genes with tendency of higher expression upon *DET1* suppression. This can be seen in the case of carotenoid biosynthesis with the downregulation of one gene (xanthine dehydrogenase/oxidase) while others showed higher expression despite not being significant. Furthermore, lycopene epsilon cyclase (*LCY*) was shown to be upregulated (Fig. 5) despite RNA-seq analysis indicated otherwise. Nonetheless, we cannot compare directly between *DET1*-suppressed papaya callus with that of tomato fruit with very different physiology and metabolism, especially in secondary metabolite biosynthesis. However, we have successfully demonstrated that *DET1* suppression exhibited similar effect on the upregulation of certain biosynthetic pathways in general.

Conclusion

In this study, embryogenic calli of *C. papaya* were successfully transformed based on *Agrobacterium*-mediated approach with RNAi silencing hpDET1 construct. The

suppression of endogenous *DET1* allowed us to identify transcriptome-wide effect and genes affected by this negative regulator of photomorphogenesis. Upregulated genes mainly involved in cell wall modification and secondary metabolite biosynthesis in contrast to downregulated genes which involved in developmental processes and response to various stimuli. This is the first proof-of-concept experiment which demonstrated that the manipulation of photoregulated pathway can increase gene expression of secondary metabolite biosynthesis in papaya callus, consistent with previous reports in tomato fruit and rapeseed. However, cellular characterisation and metabolite analysis will be required to support the effects at metabolite level as changes at transcript level do not directly translate into phenotypic changes. Therefore, characterisation of metabolite composition will be essential to ascertain the effects of *DET1* suppression on the secondary metabolites. Further study using a fruit-specific promoter in papaya is needed to avoid non-desired defect of constitutive *DET1* suppression. This study supports the possibility in manipulating light-regulated pathway as an alternative to engineering targeted biosynthetic pathways for the improvement of phytonutrient in tropical fruit crops.

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Author contributions Conceived the experiments: NDJ, NMN, HHG. Designed and performed the experiments: NDJ, ERR, HHG. Analysed the data: NDJ, HHG. Contributed reagents/materials/analysis tools: HHG. Wrote the paper: NDJ, HHG, ERR, NMN. All authors read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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